

人工智能金融期末汇报

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问题改进和模型介绍

问题改进：

回应期中“整体数据做特征工程导致数据泄露”：现改为两阶段防泄漏流程。验证阶段仅用**训练子集拟合预处理**，再对验证集做 transform，避免把验证分布（缩放参数、编码词表、分箱边界等）泄漏进特征工程；提交阶段在全量训练集重新拟合预处理并训练最终模型。保证验证指标准确性。

用到的模型

RandomForest

原理：装袋采样 + 多棵决策树平均，显著降低方差

LightGBM

原理：梯度提升，叶子优先（leaf-wise）增长，适合高维与稀疏

CatBoost

原理：Ordered Boosting，基于有序统计减少目标泄露；对类别特征天然友好

XGBoost

原理：梯度提升树；`tree_method='hist'` 直方图分裂，高效稳健

Xgboost优化

Optuna超参数优化

目标：在验证集上优化少量关键超参，结合早停提升泛化

维度：max_depth、min_child_weight、subsample、
colsample_bytree、lambda、alpha、gamma、
learning_rate

步骤：子集验证 → 早停选择 → 将最优小范围参数回写到全量训练

Bagging 多种子

做法：设定多个随机种子训练若干子模型，对测试预测做均值

作用：在树模型中显著降低方差，提升稳定性；与小规模调参（降偏差）协同

```
def _optuna_tune_xgb(X_tr_m, y_tr_t, X_val_m, y_val_t, y_val_raw):
    import optuna
    def objective(trial):
        params = {
            "max_depth": trial.suggest_int("max_depth", 6, 12),
            "min_child_weight": trial.suggest_float("min_child_weight", 1.0, 5.0),
            "subsample": trial.suggest_float("subsample", 0.5, 0.95),
            "colsample_bytree": trial.suggest_float("colsample_bytree", 0.5, 0.95),
            "reg_lambda": trial.suggest_float("reg_lambda", 0.5, 2.0),
            "reg_alpha": trial.suggest_float("reg_alpha", 0.0, 1.0),
            "gamma": trial.suggest_float("gamma", 0.0, 5.0),
            "learning_rate": trial.suggest_float("learning_rate", 0.02, 0.15),
        }
        from xgboost import XGBRegressor as _XGB
        est = _XGB(
            n_estimators=int(os.environ.get("XGB_TUNE_EST", "1200")),
            tree_method="hist",
            n_jobs=-1,
            random_state=42,
            **params,
        )
        est.set_params(eval_metric="mae")
        import xgboost as xgb
        est.fit(X_tr_m, y_tr_t, eval_set=[(X_val_m, y_val_t)],
              callbacks=[xgb.callback.EarlyStopping(
                  rounds=150, save_best=True)],)
        pred = est.predict(X_val_m)
        pred = np.expm1(pred)
        pred = np.nan_to_num(pred, nan=0.0, posinf=1e12, neginf=-1e12)
        return float(mean_absolute_error(y_val_raw, pred))
    n_trials = int(os.environ.get("OPTUNA_TRIALS", "15"))
    study = optuna.create_study(direction="minimize")
    study.optimize(objective, n_trials=n_trials, gc_after_trial=True)
    best = study.best_trial.params if hasattr(study, "best_trial") else {}
    return best
```

融合与成果

Stacking（二层融合）：用多个基础模型的验证预测作为特征，再训练一个简单的线性模型（如Ridge）进行二次学习，自动调整各模型的偏差与权重。优势是能纠正系统性误差、提升稳健性与泛化，适合模型互补的场景。

加权融合（Blended）：依据各模型在验证集上的误差（如MAE）倒数分配权重，对它们的预测做线性加权求和。优势是实现简单、计算高效、可解释性强，能快速稳定地整合多模型信息作为可靠基线。（大概能提高0.5分）

房价

Metrics	In sample rmse	out of sample rmse	Cross-validation rmse	Kaggle Sc...
RandomForest	421877.077171182...	891449.5300343917	808624.3638447054	74.8
XGBoost	570637.7924907828	1009297.13273699...	769451.8627309742	74.1
LightGBM	494351.8480770491	1015173.36578769...	670826.1194866408	74.5
CatBoost	821518.4207551314	1094812.11220851...	852226.9436430688	72.6
XGBoost_Bagg...	563326.1701247523	1037206.47979881...	665105.3683224387	74.1

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Metrics	In sample rmse	out of sample rmse	Cross-validation rmse	Kaggle Sc...
RandomForest	135191.192102278...	273568.901411001...	238849.99808210475	74.8
XGBoost	181032.672449144...	339078.135914594...	234175.05973069102	74.1
LightGBM	163393.8758654743	315924.2217138598	216243.97051794486	74.5
CatBoost	263847.460714961...	340488.99730762	270045.7773497847	72.6
XGBoost_Bagg...	178367.228307554...	351275.416097722...	215974.56048301948	74.1